

ALGORITHM:

1. Start.
2. Display the menu:
 - Count characters in a string.
 - Print each character on a new line.
 - Count vowels and consonants.
 - Exit.
3. Read the user's choice.
4. Use a switch statement:
 - Choice 1: Call countCharacters().
 - Choice 2: Call printCharacters().
 - Choice 3: Call countVowelsConsonants().
 - Choice 4: Exit the program.
 - Otherwise, display Invalid choice.
5. Repeat the menu until the user selects Exit.
6. Stop.

Function 1: Count Characters

1. Read a string.
2. Start count = 0.
3. Traverse the string until the null character '\0' is reached.
4. Increment count for every character.
5. Display the number of characters.

Function 2: Print Each Character

1. Read a string.
2. Traverse the string character by character.
3. Print every character on a separate line.

Function 3: Count Vowels and Consonants

1. Read a string.
2. Initialize vowels = 0 and consonants = 0.
3. Traverse each character.
4. If the character is a, e, i, o, u in uppercase or lowercase, increment vowels.
5. If it is any other alphabet, increment consonants.
6. Ignore spaces, digits, and special characters.
7. Display the number of vowels and consonants.

C PROGRAM:

```
#include <stdio.h>
```

```
void countCharacters();
```

```
void printCharacters();
```

```
void countVowelsConsonants();
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    do
```

```
    {
```

```
        printf("\n--- MENU ---\n");
```

```
        printf("1. Count characters in a string\n");
```

```
        printf("2. Print each character on a new line\n");
```

```
        printf("3. Count vowels and consonants\n");
```

```
        printf("4. Exit\n");
```

```
        printf("Enter your choice: ");
```

```
        scanf("%d", &choice);
```

```
getchar(); // Removes newline left by scanf()
```

```
switch (choice)
```

```
{
```

```
    case 1:
```

```
        countCharacters();
```

```
        break;
```

```
    case 2:
```

```
        printCharacters();
```

```
        break;
```

```
    case 3:
```

```
        countVowelsConsonants();
```

```
        break;
```

```
    case 4:
```

```
        printf("Exiting program...\n");
```

```
        break;
```

```
    default:
```

```
        printf("Invalid choice! Please enter 1 to 4.\n");
```

```
}
```

```
} while (choice != 4);
```

```
return 0;
```

```
}
```

```
/* Function to count characters without using strlen() */
```

```
void countCharacters()
```

```
{
```

```
    char str[100];
```

```
    int i = 0, count = 0;
```

```
    printf("Enter a string: ");
```

```
    fgets(str, sizeof(str), stdin);
```

```
    while (str[i] != '\0' && str[i] != '\n')
```

```
    {
```

```
        count++;
```

```
        i++;
```

```
    }
```

```
    printf("Number of characters = %d\n", count);
```

```
}
```

```
/* Function to print each character on a new line */
```

```
void printCharacters()
```

```
{
```

```
    char str[100];
```

```
    int i = 0;
```

```
    printf("Enter a string: ");
```

```
fgets(str, sizeof(str), stdin);
```

```
printf("Characters are:\n");
```

```
while (str[i] != '\0' && str[i] != '\n')
```

```
{
```

```
    printf("%c\n", str[i]);
```

```
    i++;
```

```
}
```

```
}
```

```
/* Function to count vowels and consonants */
```

```
void countVowelsConsonants()
```

```
{
```

```
    char str[100];
```

```
    int i = 0;
```

```
    int vowels = 0, consonants = 0;
```

```
    char ch;
```

```
    printf("Enter a string: ");
```

```
    fgets(str, sizeof(str), stdin);
```

```
    while (str[i] != '\0' && str[i] != '\n')
```

```
{
```

```
    ch = str[i];
```

```
    if ((ch >= 'A' && ch <= 'Z') ||
```

```
(ch >= 'a' && ch <= 'z'))  
{  
    if (ch == 'a' || ch == 'e' || ch == 'i' ||  
        ch == 'o' || ch == 'u' ||  
        ch == 'A' || ch == 'E' || ch == 'I' ||  
        ch == 'O' || ch == 'U')  
    {  
        vowels++;  
    }  
    else  
    {  
        consonants++;  
    }  
}  
  
    i++;  
}  
  
printf("Number of vowels = %d\n", vowels);  
printf("Number of consonants = %d\n", consonants);  
}
```